

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A

Application - Based Questions

Duration: 2Hrs

1. Design and develop a Reinforcement Learning framework a ride-sharing company uses Reinforcement Learning to match drivers with passengers efficiently. Apply RL concepts to define the state space, action space, and reward function to minimize waiting time and improve service quality. **(5 Marks)**
2. An e-learning platform uses Reinforcement Learning to recommend personalized study materials. Recall how exploration and exploitation strategies help balance familiar and new content recommendations. **(5 Marks)**
3. Design and develop a Reinforcement Learning framework a robotic vacuum cleaner uses Reinforcement Learning to clean efficiently while avoiding obstacles. Apply the concept of policy and explain how the value function improves performance over time. **(5 Marks)**
4. A cryptocurrency trading system uses Reinforcement Learning to decide buy, sell, or hold actions. Outline the challenges in designing reward functions and explain how poor reward design can affect performance. **(5 Marks)**
5. Develop a smart grid system uses Reinforcement Learning to manage electricity distribution. Create an RL model by defining state space, action space, and reward mechanism. **(5 Marks)**

6. A healthcare monitoring system uses Reinforcement Learning to adjust medication dosages. Apply RL concepts to define states, actions, and rewards, and explain how learning improves outcomes. **(5 Marks)**

7. Design a model drone delivery system uses Reinforcement Learning for navigation. Recall how exploration and exploitation strategies influence route optimization. **(5 Marks)**

8. **Design and develop** a Reinforcement Learning framework a manufacturing robot uses Reinforcement Learning to optimize assembly operations. Apply the concept of policy and explain how value functions help improve efficiency. **(5 Marks)**

9. An online advertising system uses Reinforcement Learning to maximize click-through rates. Outline reward design challenges and explain their impact on system performance. **(5 Marks)**

10. A smart irrigation system uses Reinforcement Learning to optimize water usage. Create an RL framework by defining states, actions, and reward function. **(5 Marks)**

Code of Conduct:

*I B.VANI(192472348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: B.VANI(192472348)

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

1. Reinforcement Learning for Ride-Sharing Driver-Passenger Matching

A ride-sharing company can use Reinforcement Learning (RL) to efficiently match available drivers with passengers. The RL agent observes the current transportation environment and learns actions that minimize passenger waiting time.

State Space: The state can include driver locations, passenger locations, number of available drivers, passenger demand, traffic conditions, and estimated travel time.

Action Space: The agent can assign a particular driver to a passenger, redirect a driver toward a high-demand area, or keep the driver available.

Reward Function: A positive reward can be given for successful and quick passenger matching. Negative rewards can be given for long passenger waiting times, driver idle time, and unnecessary travel distance.

Policy: The policy determines which driver should be assigned to a passenger for a given state.

Conclusion: By continuously learning from completed rides and waiting times, the system can improve driver utilization, reduce passenger waiting time, and improve overall service quality.

2. Reinforcement Learning for Personalized E-Learning Recommendations

An e-learning platform can use RL to recommend study materials according to a student's learning behavior and performance.

State: The state may include the student's previous learning history, completed topics, test scores, difficulty level, time spent studying, and current knowledge level.

Actions: The agent can recommend familiar topics, new topics, revision materials, practice questions, or advanced content.

Exploration: Exploration recommends new or less frequently selected materials to discover potentially useful content.

Exploitation: Exploitation recommends materials that are already known to provide good learning outcomes.

Conclusion: A balance between exploration and exploitation prevents the system from repeatedly recommending the same content while ensuring that effective materials continue to be recommended. This allows the system to personalize learning over time.

3. Reinforcement Learning for a Robotic Vacuum Cleaner

A robotic vacuum cleaner can use RL to learn an efficient cleaning strategy while avoiding obstacles.

State: The state can include the robot's current position, cleaned and uncleaned areas, obstacle locations, battery level, and distance from the charging station.

Actions: The robot can move forward, backward, left, right, change direction, clean an area, or return to the charging station.

Policy: The policy maps the current state to the best action. For example, if an obstacle is detected ahead, the policy may select a turning action.

Value Function: The value function estimates the expected future reward from a particular state. States that lead to more cleaning with less battery consumption receive higher values.

Conclusion: Through repeated interaction with the environment, the value function becomes more accurate and the policy improves. The robot can therefore clean more efficiently while avoiding obstacles and conserving battery power.

4. Reinforcement Learning for Cryptocurrency Trading

A cryptocurrency trading system can use RL to decide whether to buy, sell, or hold an asset.

State: The state may include cryptocurrency price, price changes, trading volume, market trends, current holdings, and available balance.

Actions: The agent can choose Buy, Sell, or Hold.

Reward: The reward can be based on the change in portfolio value after an action, while considering transaction costs and risk.

Reward Design Challenges: Designing an appropriate reward function is difficult because maximizing short-term profit may encourage excessive trading or risky decisions. If the reward ignores transaction costs, volatility, or losses, the agent may learn an undesirable strategy.

Conclusion: A well-designed reward should consider profit, risk, transaction costs, and long-term performance. Poor reward design can cause unstable trading behavior and significant financial losses.

5. Reinforcement Learning for Smart Grid Electricity Distribution

RL can be used in a smart grid to manage electricity distribution efficiently according to changing demand and supply.

State Space: The state can include electricity demand, available generation, battery charge, renewable energy production, grid load, and electricity prices.

Action Space: Actions may include allocating electricity to different areas, charging or discharging batteries, adjusting power generation, or purchasing

electricity from the grid.

Reward Mechanism: Positive rewards can be given for satisfying electricity demand efficiently and using renewable energy. Negative rewards can be assigned for energy wastage, overload, high operating cost, and unmet demand.

Policy: The policy determines the optimal electricity allocation for each grid condition.

Conclusion: The RL system can continuously adapt to changes in demand and renewable energy availability, improving energy efficiency, reducing wastage, and maintaining grid stability.

6. Reinforcement Learning for Healthcare Medication Dosage

RL can be applied to healthcare monitoring systems to assist in selecting suitable medication dosage based on a patient's observed condition. Such systems require strict medical supervision and safety constraints.

State: The state may include vital signs, symptoms, previous medication dosage, patient response, laboratory measurements, and other relevant monitoring information.

Actions: Possible actions can include maintaining the current dosage, adjusting the dosage, or requesting further monitoring. In a real system, these actions must operate within medically approved limits.

Reward: Positive rewards can represent improvement toward desired health measurements and stable patient conditions. Negative rewards can represent adverse responses, unsafe measurements, or deterioration.

Learning: The agent observes the patient's response to previous decisions and updates its policy to improve future decisions.

Conclusion: RL can support personalized treatment decisions by learning from patient responses; however, safety constraints, clinical validation, and human medical supervision are essential before practical deployment.

7. Reinforcement Learning for Drone Delivery Navigation

RL can be used to enable a delivery drone to select efficient and safe routes to its destination.

State: The state can include the drone's current location, destination, battery level, weather conditions, obstacles, altitude, and traffic or restricted zones.

Actions: The drone can change direction, altitude, speed, or select an alternative route.

Exploration: During exploration, the drone evaluates alternative routes to discover potentially shorter or safer paths.

Exploitation: During exploitation, it uses the best-known route based on previous experience.

Conclusion: Balancing exploration and exploitation helps the drone discover

efficient routes while continuing to use routes that have already demonstrated good performance. Safety restrictions must always override unsafe exploratory actions.

8. Reinforcement Learning for Manufacturing Robot Assembly

RL can optimize the sequence of operations performed by a manufacturing robot during assembly.

State: The state may include the robot's current position, assembly stage, component locations, tool status, remaining components, and machine condition.

Actions: The robot can select the next assembly operation, move to a component, change a tool, or adjust its movement.

Policy: The policy determines the best assembly action for each state. An improved policy can reduce unnecessary movements and assembly time.

Value Function: The value function estimates the expected future reward from a state. States leading to faster and accurate assembly receive higher values.

Reward: Positive rewards can be given for successful assembly and reduced completion time, while penalties can be given for errors, collisions, damaged components, and unnecessary movements.

Conclusion: Through repeated learning, the robot can develop an efficient assembly policy, reduce production time, minimize errors, and improve resource utilization.

9. Reinforcement Learning for Online Advertising

RL can be used in an online advertising system to select advertisements that maximize click-through rates (CTR).

State: The state may include user interests, browsing history, previous advertisement interactions, device type, time, location information, and current webpage context.

Actions: The agent can select which advertisement to display or choose not to display an advertisement.

Reward: A positive reward can be assigned when users interact with an advertisement, while additional rewards may be based on meaningful conversions. Negative or zero rewards can be assigned when advertisements are ignored.

Reward Design Challenges: Optimizing only CTR may encourage the system to select advertisements that receive clicks but do not produce meaningful engagement. It may also encourage excessive advertising.

Impact: An improperly designed reward can reduce user satisfaction and lead to poor long-term performance. Therefore, the reward should consider CTR,

conversions, user experience, and long-term engagement.

Conclusion: A carefully designed reward function allows RL to maximize advertising effectiveness while maintaining a positive user experience.

10. Reinforcement Learning for Smart Irrigation

RL can optimize irrigation by determining when and how much water should be supplied to crops.

State Space: The state can include soil moisture, temperature, humidity, rainfall forecast, crop growth stage, water availability, and current weather conditions.

Action Space: The agent can choose the amount of water to supply, irrigation duration, irrigation time, or choose not to irrigate.

Reward Function: Positive rewards can be given for maintaining suitable soil moisture and healthy crop conditions. Negative rewards can be assigned for excessive water usage, insufficient irrigation, and crop stress.

Policy: The policy determines the appropriate irrigation action for each environmental condition.

Conclusion: By continuously learning from soil sensors, weather information, and crop responses, the RL system can adjust irrigation dynamically. This can reduce water wastage, maintain suitable soil moisture, and improve irrigation efficiency.